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Practical Exam Tasks

1. Create a directory structure as shown below and show it in the terminal.

```
gajur@LAPTOP-709700PC: ~  
gajur@LAPTOP-709700PC:~$ mkdir -p Dev/projects Dev/scripts Dev/backups Dev/logs  
gajur@LAPTOP-709700PC:~$ tree  
├── Dev  
│   ├── backups  
│   ├── logs  
│   ├── projects  
│   └── scripts  
6 directories, 0 files
```

Figure 1: Creating a Directory

2. Inside projects/, create two subdirectories alpha and beta using a relative path.

```
gajur@LAPTOP-709700PC:~/Dev/projects$ cd ..  
gajur@LAPTOP-709700PC:~/Dev$ mkdir -p projects/alpha projects/beta  
gajur@LAPTOP-709700PC:~/Dev$ tree  
├── backups  
├── logs  
├── projects  
│   ├── alpha  
│   └── beta  
└── scripts  
7 directories, 0 files
```

Figure 2: Inside Projects creating two subdirectories

3. Create an empty file readme.md in alpha/ using an absolute path.

```
gajur@LAPTOP-709700PC:~$ touch /home/gajur/Dev/projects/alpha/readme.md  
gajur@LAPTOP-709700PC:~$ cd Dev/projects/alpha  
gajur@LAPTOP-709700PC:~/Dev/projects/alpha$ ls  
readme.md  
gajur@LAPTOP-709700PC:~/Dev/projects/alpha$ tree  
├── readme.md  
1 directory, 1 file  
gajur@LAPTOP-709700PC:~/Dev/projects/alpha$ |
```

Figure 3: Creating empty file readme.md in alpha

4. Use the touch command with an absolute path to create a file named deploy.sh in the scripts directory.

```
gajur@LAPTOP-709700PC:~$ touch /home/gajur/Dev/scripts/deploy.sh  
gajur@LAPTOP-709700PC:~$ cd Dev/scripts  
gajur@LAPTOP-709700PC:~/Dev/scripts$ ls  
deploy.sh
```

Figure 4: Creating file deploy.sh in scripts directory

5. Display the detailed permissions and ownership of the deploy.sh file.

```
gajur@LAPTOP-709700PC:~/Dev/scripts$ ls -l  
total 0  
-rw-r--r-- 1 gajur gajur 0 Jan 20 07:32 deploy.sh
```

Figure 5: Displaying detailed permissions

6. Set the permissions on deploy.sh so that the owner has rwx permissions and others have not.

```
gajur@LAPTOP-709788P0:~/Dev/scripts$ chmod u+rwx,g-rwx,o-rwx deploy.sh
gajur@LAPTOP-709788P0:~/Dev/scripts$ ls -l
total 8
-rwx----- 1 gajur gajur 0 Jan 20 07:32 deploy.sh
```

Figure 6: Set permissions on owner

7. Set permissions of beta to:

- Users: RWX
- Group: RX
- Others: None

```
gajur@LAPTOP-709788P0:~/Dev/projects$ chmod u=rwx,g=rx,o= beta
gajur@LAPTOP-709788P0:~/Dev/projects$ ls -l
total 8
drwxr-xr-x 2 gajur gajur 4096 Jan 20 07:27 alpha
drwxr-x--- 2 gajur gajur 4096 Jan 20 08:10 beta
```

Figure 7: Set permissions on beta

8. Copy alpha from projects/ to logs/.

```
gajur@LAPTOP-709788P0:~$ cp -r Dev/projects/alpha Dev/logs
gajur@LAPTOP-709788P0:~$ tree
├── Dev
│   ├── backups
│   ├── logs
│   │   ├── alpha
│   │   └── readme.md
│   ├── projects
│   │   ├── alpha
│   │   ├── beta
│   │   └── scripts
│   │       └── deploy.sh
└── 9 directories, 3 files
```

Figure 8: Copy alpha from projects to logs

9. Rename the copied file in logs/ to alpha_old.sh

```
gajur@LAPTOP-709788P0:~$ mv Dev/logs/alpha/readme.md alpha_old.sh
gajur@LAPTOP-709788P0:~$ tree
├── Dev
│   ├── backups
│   ├── logs
│   │   ├── alpha
│   │   └── alpha_old.sh
│   ├── projects
│   │   ├── alpha
│   │   ├── beta
│   │   └── scripts
│   │       └── deploy.sh
└── 9 directories, 3 files
```

Figure 9: Renamed copied file in logs to alpha_old.sh

10. Delete the beta/ directory from projects/.

```
gajur@LAPTOP-709788P0:~$ rmdir Dev/projects/beta
gajur@LAPTOP-709788P0:~$ tree
.
├── Dev
│   ├── backups
│   ├── logs
│   │   ├── alpha
│   │   └── alpha_old.sh
│   ├── projects
│   │   ├── alpha
│   │   └── readme.md
│   └── scripts
│       └── deploy.sh
└── 8 directories, 3 files
```

Figure 10: Delete beta directory from projects